

ECON 101—Introduction to Microeconomics
Fall 2019

Midterm I

September 23, 2019

ANSWER KEY

Part I: Multiple Choice

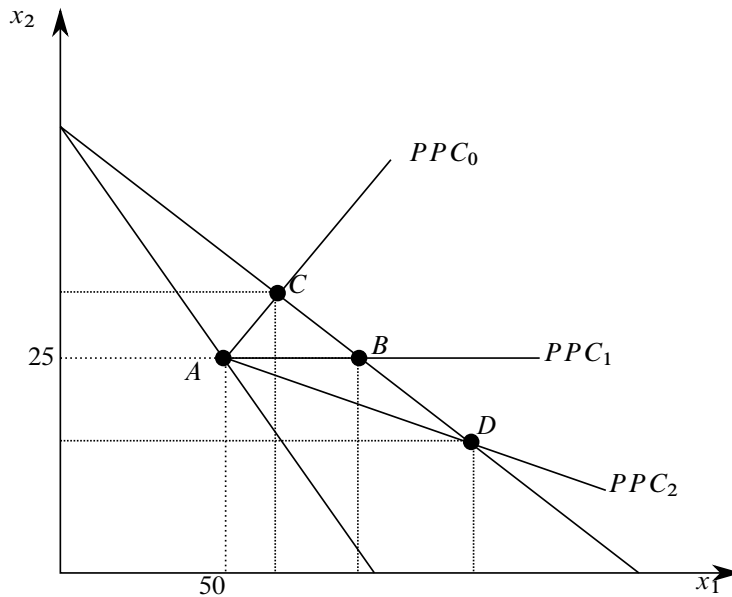
Please record all questions in the answer sheet. Each question is worth 2 points for a total of 40 points.

Version	I	II	III	IV	V
Question					
1	(d)	(d)	(b)	(d)	(b)
2	(c)	(a)	(d)	(a)	(a)
3	(a)	(c)	(d)	(a)	(b)
4	(b)	(b)	(c)	(a)	(a)
5	(b)	(a)	(a)	(d)	(c)
6	(c)	(c)	(c)	(d)	(d)
7	(c)	(a)	(b)	(d)	(a)
8	(c)	(b)	(d)	(a)	(c)
9	(a)	(d)	(c)	(a)	(a)
10	(a)	(c)	(a)	(d)	(c)
11	(d)	(b)	(a)	(a)	(d)
12	(a)	(d)	(a)	(a)	(d)
13	(a)	(b)	(d)	(b)	(c)
14	(d)	(c)	(a)	(a)	(b)
15	(c)	(b)	(a)	(a)	(d)
16	(b)	(a)	(c)	(b)	(b)
17	(b)	(d)	(b)	(d)	(a)
18	(a)	(a)	(c)	(b)	(a)
19	(b)	(a)	(a)	(b)	(c)
20	(a)	(b)	(d)	(a)	(c)

Part II: Open problems

Instructions: Write your answers in the space provided in the answer sheet.

1. Consider the following scenario as depicted by the following figure.



There are three consumers, denoted by $i = 0, 1, 2$ which can consume two goods, x_1 and x_2 . The consumers have the same Income of \$100 dollars. The price of x_1 is \$1 dollar and the price of x_2 is \$2. Each consumer initially chooses the initial consumption bundle $x_1 = 50, x_2 = 25$ which is denoted by point A. Suppose that there is a reduction in the price of x_1 to \$0.5. After this price decrease, the optimal consumption bundle of consumer 0 is at point C, the optimal bundle of consumer 1 is at point B and the optimal bundle of consumer 2 is at point D. In the graph you can see the depiction of their *price consumption curves*, denoted by PPC_i for $i = 0, 1, 2$.

Suppose that a market researcher tells you that after estimating the own price elasticities (arc) for the demand of good x_1 for each consumer, he has found the following:

Unfortunately, the market researcher has been very careless and he forgot to record which elasticity (arc) corresponds to each consumer.

- (5 points) With the data above, match each elasticity with its corresponding consumer and explain why?
 - (10 points) What does the own price elasticity and the slope of the PPC curves tell us about the cross-price elasticity of good x_2 with respect to p_1 ?
 - (15 points) Using the previous information, compute the optimal consumption bundles C, B and D of each consumer.
2. Suppose that an economy has two individuals, Alejandro and Oleg. Alejandro's demand curve is given by $q_D^A = 100,000 - p$ and Oleg's demand curve is given by $q_D^O = 20,000 - 0.5P$. Furthermore, there are initially 1,000 identical firms with the following cost structure:

Firm's Output	1	2	3	4	5
MC	11	11.13	12.00	13.63	16.00
ATC	13.50	12.25	12.00	12.19	12.70
AVC	11.25	11.13	11.25	11.63	12.25

- (6 points) Perform horizontal summation of these demand curves to find the market demand curve. Denote this curve by Q_D
- (6 points) What is the firm's shutdown point and what is the firm's supply schedule?
- (6 points) What is the market price and quantity?
- (6 points) Will firms enter or exit?
- (6 points) Calculate the long-run equilibrium price, quantity, and number of firms.